

# DD\_REP-01 - Reporting Data Mart and Dashboard API

Version: v1.0

Status: Draft for review

Bundle: 10\_other

Depends on:

- DD\_CROSS-00-System\_Traceability\_and\_Implementation\_Map-v1.0
- FOUNDATION\_AUTH
- FOUNDATION\_KAFKA
- FOUNDATION\_DB
- DD\_SALES-01-Lead\_Territory\_and\_Franchise\_Sales\_Management-v1.0
- DD\_SUB-LM-01-Subscription\_Lifecycle\_Management-v1.0
- DD\_SUB-WF-FRAMEWORK-01-Subscription\_Workflow\_Framework-v1.0
- DD\_BIL-01-Charging\_Engine-Core-v1.1
- DD\_BIL-02-Invoicing-Core-v2.11
- DD\_BIL-04-Dunning\_Engine-v1.4
- DD\_FUL-02-Order\_Capture-v1.0
- DD\_WO-01-Work\_Order\_Module-v1.0
- DD\_OSR-01-Stock\_Chain\_Engine-v1.0
- DD\_TCK-01-Ticketing\_and\_Case\_Management-v1.0

MVP simplification baseline: REP-01 is mandatory for dashboards, exports, and aggregate reporting. First release scope is a small mart for revenue, payments, active subscriptions, activations, tickets, work orders, stock summary, and dunning. Do not let reporting screens fan out to operational modules for aggregate totals.

## 1. Purpose

This DD defines the reporting backend used by FE-APP-05 Reporting Portal and embedded operational dashboards in Backoffice.

REP-01 prevents reporting screens from directly querying operational module databases. It builds read models/data marts from module events, controlled extracts, and approved read APIs.

## 2. Scope

Included:

| Capability      | Purpose                                                   |
|-----------------|-----------------------------------------------------------|
| Event ingestion | Consume module events into reporting staging tables.      |
| Data mart       | Build daily and near-real-time aggregates for dashboards. |
| Dashboard APIs  | Provide stable APIs for FE-APP-05 and Backoffice widgets. |
| Export controls | Controlled CSV/XLS/PDF exports with audit.                |
| Reconciliation  | Compare reporting aggregates to source module counts.     |
| PII masking     | Enforce role-based masking for reports.                   |

Out of scope:

| Out of scope                                 | Owner                                   |
|----------------------------------------------|-----------------------------------------|
| Source module transactions                   | Owning operational module.              |
| Ad hoc direct SQL into production module DBs | Not allowed.                            |
| Payroll/commission payout                    | Future finance/commission DD if needed. |

|                            |                                                              |
|----------------------------|--------------------------------------------------------------|
| Out of scope               | Owner                                                        |
| BI vendor dashboard design | Implementation choice, but APIs/data mart rules still apply. |

### 3. Data Sources

| Source module | Data needed                                              | Method                                    |
|---------------|----------------------------------------------------------|-------------------------------------------|
| SALES-01      | Lead funnel, attribution, territory, agent productivity. | Kafka events and daily extract if needed. |
| ILM           | Customer/account counts and KYC state.                   | Events/API extract with PII masking.      |
| SUB-LM/SUB-WF | Subscription base, lifecycle operations, MACD outcomes.  | Events plus periodic reconciliation API.  |
| BIL           | Revenue, invoices, payments, wallet, dunning.            | Billing events and daily summary extract. |
| FUL           | Order funnel and activation pipeline.                    | Order/fulfillment events.                 |
| WO            | Work order backlog, assignment, SLA, finalization.       | WO events.                                |
| OSR           | Stock levels, movements, swaps/RMA.                      | OSR events and stock snapshot extract.    |
| TCK           | Ticket volumes, SLA, backlog, WO-linked tickets.         | TCK events.                               |
| NOT/ICN       | Notification delivery and staff task metrics.            | Notification events.                      |

Rule: reporting may store copied/derived data, but operational modules remain source of truth.

### 4. Architecture

Recommended medium deployment:

```
Module outbox events -> Kafka -> REP ingestion workers -> reporting_staging
                                                    -> aggregate jobs -> reporting_mart
FE-APP-05 / Backoffice -> REP dashboard APIs -> reporting_mart
```

For budget-conscious deployment, REP can start on the same PostgreSQL cluster as other non-critical modules but should use a separate database/schema. For production growth, move reporting to a separate read-optimized PostgreSQL/warehouse instance.

### 5. Core Tables

#### 5.1 rep\_event\_ingestion

What it is for: records consumed events and ingestion state.

When to use it: every Kafka event consumed by reporting.

| Column        | Type      | Purpose                     | Sample               |
|---------------|-----------|-----------------------------|----------------------|
| event_id      | TEXT PK   | Source event id.            | evt_01J9...          |
| source_module | TEXT      | Producer.                   | BIL-02               |
| event_type    | TEXT      | Event name.                 | InvoiceIssued        |
| operator_code | TEXT      | Operator.                   | WIK                  |
| occurred_at   | TIMESTAMP | Source event time.          | 2026-05-28T10:00:00Z |
| processed_at  | TIMESTAMP | REP processing time.        | 2026-05-28T10:00:03Z |
| status        | TEXT      | PROCESSED, FAILED, SKIPPED. | PROCESSED            |
| payload_hash  | TEXT      | Change detection.           | sha256:...           |

#### 5.2 rep\_daily\_metric

What it is for: stores daily aggregate metrics by operator and dimensions.

When to use it: dashboard cards, trends, exports.

| Column        | Type | Purpose         | Sample     |
|---------------|------|-----------------|------------|
| metric_date   | DATE | Reporting date. | 2026-05-28 |
| operator_code | TEXT | Operator.       | WIK        |

| Column         | Type      | Purpose           | Sample                                |
|----------------|-----------|-------------------|---------------------------------------|
| metric_code    | TEXT      | Metric name.      | orders_activated_count                |
| dimension_json | JSONB     | Dimensions.       | {"territory": "KE-NRB-LAVINGT<br>ON"} |
| metric_value   | NUMERIC   | Value.            | 42                                    |
| computed_at    | TIMESTAMP | Computation time. | 2026-05-29T00:10:00Z                  |

### 5.3 rep\_dashboard\_snapshot

What it is for: stores precomputed dashboard payloads where queries are expensive.

When to use it: management dashboards, finance dashboards, high-traffic summary pages.

| Column         | Type      | Purpose          | Sample               |
|----------------|-----------|------------------|----------------------|
| snapshot_id    | UUID PK   | Snapshot id.     | snap_01J...          |
| dashboard_code | TEXT      | Dashboard.       | sales_daily_overview |
| operator_code  | TEXT      | Operator.        | WIK                  |
| period_start   | DATE      | Start date.      | 2026-05-01           |
| period_end     | DATE      | End date.        | 2026-05-28           |
| payload_json   | JSONB     | Dashboard data.  | {"cards": [...]}     |
| generated_at   | TIMESTAMP | Generation time. | 2026-05-28T18:00:00Z |

### 5.4 rep\_export\_job

What it is for: tracks long-running exports and audit.

When to use it: a user downloads CSV/XLS/PDF report.

| Column               | Type      | Purpose                                   | Sample               |
|----------------------|-----------|-------------------------------------------|----------------------|
| export_job_id        | UUID PK   | Export job.                               | exp_01J...           |
| dashboard_code       | TEXT      | Report source.                            | billing_aging        |
| requested_by_user_id | TEXT      | Auth user.                                | kc_user_finance_3    |
| operator_code        | TEXT      | Operator.                                 | WIK                  |
| status               | TEXT      | PENDING, RUNNING, READY, FAILED, EXPIRED. | READY                |
| file_ref             | TEXT NULL | File storage reference.                   | file_rep_01J...      |
| created_at           | TIMESTAMP | Request time.                             | 2026-05-28T11:00:00Z |
| expires_at           | TIMESTAMP | Expiry.                                   | 2026-06-04T11:00:00Z |

## 6. Dashboard Domains

| Dashboard domain  | Metrics                                                                                  | Source modules     |
|-------------------|------------------------------------------------------------------------------------------|--------------------|
| Sales funnel      | Leads created, qualified, converted, lost, conversion rate by agent/franchise/territory. | SALES, FUL, ILM.   |
| Order fulfillment | Orders by phase, KYC SLA, payment wait, install wait, activation success/failure.        | FUL, BIL, WO, SUB. |
| Subscription base | Active/paused/restricted/suspended/terminated subscriptions, MACD counts.                | SUB-LM, SUB-WF.    |
| Billing           | Invoices issued, paid, overdue, revenue, adjustments, wallet balances.                   | BIL.               |
| Dunning           | Dunning levels, suspension candidates, cured accounts, termination review.               | BIL-04, SUB-WF.    |
| Work order        | Backlog, assignment SLA, completion rate, contractor performance.                        | WO, EM.            |
| Equipment         | Stock by location, movements, swaps, RMA, recovery exceptions.                           | OSR.               |
| Ticketing         | Open tickets, SLA breach, category mix, WO-linked tickets.                               | TCK, WO.           |

| Dashboard domain | Metrics                                               | Source modules |
|------------------|-------------------------------------------------------|----------------|
| Notifications    | Delivery success/failure, staff task acknowledgement. | NOT, ICN.      |

## 7. APIs

All APIs require Authorization: Bearer <jwt> and role guards.

| API                                                                                   | Purpose                           |
|---------------------------------------------------------------------------------------|-----------------------------------|
| GET /api/reports/dashboards                                                           | List dashboards allowed for user. |
| GET /api/reports/dashboards/{dashboard_code}?operatorCode=WIK&period=TODAY            | Dashboard payload.                |
| GET /api/reports/metrics/{metric_code}?operatorCode=WIK&from=2026-05-01&to=2026-05-28 | Metric time series.               |
| POST /api/reports/exports                                                             | Start export job.                 |
| GET /api/reports/exports/{export_job_id}                                              | Export status.                    |
| GET /api/reports/exports/{export_job_id}/download                                     | Authorized export download.       |
| POST /api/reports/reconcile/{source_module}                                           | Run reconciliation job.           |
| GET /api/reports/reconcile/{run_id}                                                   | Reconciliation result.            |

## Dashboard Response Example

```
{
  "dashboardCode": "sales_daily_overview",
  "operatorCode": "WIK",
  "period": {
    "from": "2026-05-28",
    "to": "2026-05-28",
    "timezone": "Africa/Nairobi"
  },
  "cards": [
    {
      "metricCode": "leads_created_count",
      "labelKey": "reports.sales.leadsCreated",
      "value": 128,
      "trend": { "direction": "UP", "percent": 12.4 }
    }
  ],
  "generatedAt": "2026-05-28T18:00:00Z",
  "sourceFreshness": [
    { "module": "SALES-01", "lastEventAt": "2026-05-28T17:59:40Z" }
  ]
}
```

## 8. Security And Privacy

| Rule      | Description                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------|
| REP-SEC-1 | FOUNDATION_AUTH roles control dashboard visibility.                                                |
| REP-SEC-2 | Exports require explicit permission and audit row.                                                 |
| REP-SEC-3 | PII fields must be masked unless the user has an approved role.                                    |
| REP-SEC-4 | Export files use FOUNDATION_FILE_STORAGE and expire.                                               |
| REP-SEC-5 | Report APIs must always scope by operator unless user is explicitly multi-operator.                |
| REP-SEC-6 | Customer-level drill-down must link back to owning module UI/API; do not expose hidden copied PII. |

## 9. Reconciliation

Reporting data must be trusted enough for management, but source modules remain authoritative.

Required reconciliations:

| Reconciliation       | Method                                                        |
|----------------------|---------------------------------------------------------------|
| Invoice count/value  | Compare REP daily invoice totals to BIL summary API.          |
| Payment totals       | Compare REP payment totals to BIL payment summary API.        |
| Active subscriptions | Compare REP active count to SUB-LM count API.                 |
| WO backlog           | Compare REP backlog to WO list/count API.                     |
| Lead conversion      | Compare SALES converted leads to FUL order attribution count. |
| Ticket backlog       | Compare REP open tickets to TCK open ticket count.            |

## 10. Frontend Mapping

| FE-APP-05 screen    | REP APIs                                         | Drill-down owner                       |
|---------------------|--------------------------------------------------|----------------------------------------|
| Sales dashboard     | GET /api/reports/dashboards/sales_daily_overview | SALES lead screens, FUL order screens. |
| Order funnel        | GET /api/reports/dashboards/order_funnel         | FUL order monitor.                     |
| Billing dashboard   | GET /api/reports/dashboards/billing_overview     | BIL invoice/payment screens.           |
| Dunning dashboard   | GET /api/reports/dashboards/dunning_overview     | BIL-04 queues, SUB-WF operations.      |
| WO dashboard        | GET /api/reports/dashboards/work_order_overview  | WO dispatch/detail.                    |
| Equipment dashboard | GET /api/reports/dashboards/equipment_overview   | OSR stock/equipment screens.           |
| Ticket dashboard    | GET /api/reports/dashboards/ticketing_overview   | TCK ticket detail.                     |
| Exports             | POST /api/reports/exports, download API.         | File storage.                          |

## 11. Infra Recommendation

Budget-conscious medium deployment:

| Component        | Recommendation                                                                        |
|------------------|---------------------------------------------------------------------------------------|
| Ingestion worker | 2 pods, 1-2 vCPU, 2-4 GB RAM.                                                         |
| API pods         | 2 pods, 1-2 vCPU, 2-4 GB RAM.                                                         |
| Database         | Separate reporting DB/schema; start PostgreSQL, partition large fact tables by month. |
| Storage          | Use file storage for exports.                                                         |
| Cache            | Redis for dashboard payload cache.                                                    |

Growth path:

- move reporting DB to separate managed PostgreSQL/read-optimized warehouse;
- add columnar store only if PostgreSQL aggregates become insufficient;
- keep event ingestion idempotent so backfill/replay is possible.

## 12. Implementation Notes

Start with:

1. Create rep\_event\_ingestion, rep\_daily\_metric, rep\_dashboard\_snapshot, and rep\_export\_job.
2. Consume key events from SALES, FUL, BIL, SUB, WO, OSR, TCK.
3. Implement sales/order/billing/WO/ticket daily metrics first.
4. Implement dashboard list/detail APIs.
5. Implement export jobs with file storage.
6. Add reconciliation jobs for BIL, SUB, WO, TCK.

7. Build FE-APP-05 against REP APIs only.

## 13. Acceptance Checklist

- ☐ FE-APP-05 does not query operational module DBs.
- ☐ Dashboard APIs are role and operator scoped.
- ☐ At least five core module event streams are ingested idempotently.
- ☐ Exports are audited and expire.
- ☐ Reconciliation exists for billing and subscription counts.
- ☐ PII masking is enforced by role.